

ECE 313: Problem Set 8: Problems and Solutions

Due: Friday, April 4 at 7:00:00 p.m.

Reading: *ECE 313 Course Notes*, Section 3.7 - 3.8

Note on reading: For most sections of the course notes there are short answer questions at the end of the chapter. We recommend that after reading each section you try answering the short answer questions. Do not hand these in; answers to the short answer questions are provided in the appendix of the notes.

Note on turning in homework: Homework is assigned on a weekly basis on Fridays, and is due by 7 p.m. on the following Friday. **Please write down your work and derivations. An answer without justification as of how it is found will not be accepted.** You must upload handwritten homework to Gradescope. Alternatively, you can typeset the homework in LaTeX. However, no additional credit will be awarded to typeset submissions. No late homework will be accepted.

Please write on the top right corner of the first page:

NAME

NETID

SECTION

PROBLEM SET #

Page numbers are encouraged but not required. Five points will be deducted for improper headings. Please assign your uploaded pages to their respective question numbers while submitting your homework on Gradescope. **5 points will be deducted for incorrectly assigned page numbers.**

1. **[ML Parameter Estimation]**

Calls arrive in a call center according to a Poisson process with arrival rate λ (calls/hour). Derive the maximum likelihood estimate $\hat{\lambda}_{ML}$ if k calls are received from 1:00 PM to 1:15 PM.

Solution: The time interval is 15 minutes, i.e. $\frac{1}{4}$ hours, so

$$P = \frac{(\lambda/4)^k}{k!} e^{-\lambda/4}$$

Then

$$\begin{aligned} \frac{\partial P}{\partial \lambda} &= \frac{k \cdot (\lambda/4)^{k-1}}{4 \cdot k!} e^{-\lambda/4} - \frac{(\lambda/4)^k}{4 \cdot k!} e^{-\lambda/4} \\ &= \frac{(\lambda/4)^{k-1}}{4 \cdot k!} e^{-\lambda/4} \left(k - \frac{\lambda}{4} \right) \end{aligned}$$

Setting this derivative to 0 results in $\hat{\lambda}_{ML} = 4k$.

2. **[Function of a Uniform Random Variable]**

Let X be a uniform random variable with mean m and variance $\frac{4m^2}{3}$, where m is a positive constant. Find the pdf of $Y = |X|$.

Solution: Assume $X \sim Unif[a, b]$, then

$$\begin{aligned} a + b &= 2m \\ (b - a)^2 &= 16m^2 \end{aligned}$$

which yields $a = -m$, $b = 3m$. The support of Y is then $[0, 3m]$. If $0 \leq c \leq m$, $P(Y \leq c) = P(-c \leq X \leq c) = \frac{c}{2m}$; if $m < c \leq 3m$, $P(Y \leq c) = P(-m \leq X \leq c) = \frac{c+m}{4m}$. Therefore,

$$F_Y(c) = \begin{cases} 0, & \text{if } c < 0, \\ \frac{c}{2m}, & \text{if } 0 \leq c \leq m, \\ \frac{c+m}{4m}, & \text{if } m < c \leq 3m, \\ 1, & \text{if } c > 3m \end{cases}$$

and thus

$$f_Y(v) = \begin{cases} \frac{1}{2m}, & \text{if } 0 \leq c \leq m, \\ \frac{1}{4m}, & \text{if } m < c \leq 3m, \\ 0, & \text{otherwise.} \end{cases}$$

3. [Function of Gaussian Random Variable]

Suppose X is a Gaussian random variable with mean 0 and variance 4.

- (a) Let $Y = \alpha$ if $X > 0$ and $Y = -\alpha$ otherwise. Does Y have a pmf or pdf? What is the pmf or pdf of Y ?

Solution: The support of Y is $\{\alpha, -\alpha\}$, and is therefore a discrete random variable, since it has finitely many outcomes. The pmf is

$$\begin{aligned} p_Y(y) &= \begin{cases} P(X > 0), & \text{if } y = \alpha, \\ P(X \leq 0), & \text{if } y = -\alpha, \\ 0, & \text{otherwise} \end{cases} \\ &= \begin{cases} \frac{1}{2}, & \text{if } y = \alpha \text{ or } y = -\alpha, \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

- (b) The squared error in representing X by Y (as defined in (a)) is defined as the random variable $Z = \begin{cases} (X - \alpha)^2, & \text{if } X > 0, \\ (X + \alpha)^2, & \text{if } X \leq 0 \end{cases}$. Find the value of α to minimize $E[Z]$.

Solution:

$$\begin{aligned} E[Z] &= \int_{-\infty}^0 (x + \alpha)^2 f_X(x) dx + \int_0^{\infty} (x - \alpha)^2 f_X(x) dx \\ &= \int_{-\infty}^{\infty} (x^2 + \alpha^2) f_X(x) dx + 2\alpha \int_{-\infty}^0 x f_X(x) dx - 2\alpha \int_0^{\infty} x f_X(x) dx \\ &= \int_{-\infty}^{\infty} (x^2 + \alpha^2) f_X(x) dx + 2\alpha \int_{-\infty}^0 x f_X(x) dx + 2\alpha \int_{-\infty}^0 x f_X(x) dx \\ &= E[X^2] + \alpha^2 + \frac{2\alpha}{\sqrt{2\pi}} \int_{-\infty}^0 x e^{-\frac{x^2}{8}} dx \end{aligned}$$

After u -substitution,

$$\begin{aligned} &= E[X^2] + \alpha^2 + \frac{2\alpha}{\sqrt{2\pi}} \int_0^1 -4du \\ &= 4 + \alpha^2 - \frac{8\alpha}{\sqrt{2\pi}} \end{aligned}$$

This is a convex function of α and is minimized at $\alpha = \frac{4}{\sqrt{2\pi}}$.

(Note: With this formulation, $E[Z]$ is the the mean squared error of Y .)

- (c) Let W be the value in $\{-4, -2, 0, 2, 4\}$ that is closest to X . That is, $W = -4$ if $X < -3$, $W = -2$ if $-3 \leq X < -1$, etc. Find the pmf (or pdf) of W . Express your answer in terms of Q -functions if necessary.

Solution: W is a discrete random variable with pmf

$$p_W(w) = \begin{cases} P(X < -3) = Q(\frac{3}{2}), & \text{if } w = -4, \\ P(-3 \leq X < -1) = Q(\frac{1}{2}) - Q(\frac{3}{2}), & \text{if } w = -2, \\ P(-1 \leq X < 1) = 1 - 2Q(\frac{1}{2}), & \text{if } w = 0, \\ P(1 \leq X < 3) = Q(\frac{1}{2}) - Q(\frac{3}{2}), & \text{if } w = 2, \\ P(X \geq 3) = Q(\frac{3}{2}), & \text{if } w = 4, \\ 0, & \text{otherwise} \end{cases}$$

4. [Function of Laplacian Random Variable]

Suppose X is a random variable with pdf $f_X(x) = \frac{1}{4}e^{-|x|/2}$, $x \in \mathbb{R}$.

- (a) Solve 3(a) for this new definition of X .

Solution: $f_X(x)$ is an even function, so $P(X > 0) = P(X \leq 0) = \frac{1}{2}$. Therefore, the pmf of Y is again:

$$p_Y(y) = \begin{cases} \frac{1}{2}, & \text{if } y = \alpha \text{ or } y = -\alpha, \\ 0, & \text{otherwise} \end{cases}$$

- (b) Solve 3(b) for this new definition of X . (**Hint:** Using integration by parts, $\int x^2 e^{-\frac{x}{2}} dx = -e^{-\frac{x}{2}}(2x^2 + 8x + 16) + C$, and $\int x e^{-\frac{x}{2}} dx = -2e^{-\frac{x}{2}}(x + 2) + C$)

Solution:

$$\begin{aligned} E[Z] &= \int_{-\infty}^0 (x + \alpha)^2 f_X(x) dx + \int_0^{\infty} (x - \alpha)^2 f_X(x) dx \\ &= \frac{1}{4} \left(\int_{-\infty}^0 (x + \alpha)^2 e^{\frac{x}{2}} dx + \int_0^{\infty} (x - \alpha)^2 e^{-\frac{x}{2}} dx \right) \\ &= \frac{1}{4} \left(\int_0^{\infty} (x' - \alpha)^2 e^{-\frac{x'}{2}} dx' + \int_0^{\infty} (x - \alpha)^2 e^{-\frac{x}{2}} dx \right) \\ &= \frac{1}{2} \int_0^{\infty} (x - \alpha)^2 e^{-\frac{x}{2}} dx \\ &= \frac{1}{2} \left(\left[-e^{-\frac{x}{2}}(2x^2 + 8x + 16) \right]_0^{\infty} - 2\alpha \left[-2e^{-\frac{x}{2}}(x + 2) \right]_0^{\infty} + \alpha^2 \left[-2e^{-\frac{x}{2}} \right]_0^{\infty} \right) \\ &= 8 - 4\alpha + \alpha^2 \end{aligned}$$

This is a convex function of α and is minimized at $\alpha = 2$.

5. [Generating a Random Variable]

Find a function g such that if U is uniformly distributed over $[0, 1]$, $Y = g(U)$ has the following pdf:

$$f_Y(v) = \begin{cases} -\frac{v+2}{4}, & \text{if } -4 \leq v < -2, \\ \frac{v+2}{4}, & \text{if } -2 \leq v < 0, \\ 0, & \text{otherwise.} \end{cases}$$

Solution: The CDF of Y is

$$F_Y(c) = \begin{cases} 0, & \text{if } c < -4, \\ -\frac{(c+2)^2}{8} + \frac{1}{2}, & \text{if } -4 \leq c < -2, \\ \frac{(c+2)^2}{8} + \frac{1}{2}, & \text{if } -2 \leq c < 0, \\ 1, & \text{if } c \geq 0 \end{cases}$$

When $-4 \leq c < -2$, $0 \leq F_Y(c) < \frac{1}{2}$, and when $-2 \leq c < 0$, $\frac{1}{2} \leq F_Y(c) < 1$.

$$g(u) = F_Y^{-1}(u) = \begin{cases} -\sqrt{4-8u} - 2, & \text{if } 0 \leq u < \frac{1}{2}, \\ \sqrt{8u-4} - 2, & \text{if } \frac{1}{2} \leq u < 1, \\ 0, & \text{otherwise.} \end{cases}$$